

1. Create one IAM user and assign EC2 and S3 full access roles.

- ➔ Open the AWS Management Console and navigate to IAM.
- ➔ Go to Users → Create user.
- ➔ Enter a username.
- ➔ Enable AWS Management Console access if required.
- ➔ Create the user.
- ➔ Open the created user and go to the Permissions tab.
- ➔ Click Add permissions → Add permissions directly.
- ➔ Attach the AmazonEC2FullAccess policy.
- ➔ Attach the AmazonS3FullAccess policy.
- ➔ Review and save the permissions.
- ➔ Verify that the user has full access to EC2 and S3 services

The screenshot shows the 'Specify user details' step of the AWS IAM 'Create user' process. A green notification bar at the top states 'User "Test_CLI" deleted.' The left sidebar shows a progress indicator with four steps: 'Specify user details' (selected), 'Set permissions', 'Review and create', and 'Retrieve password'. The main content area is titled 'Specify user details' and contains a 'User details' section. In this section, the 'User name' is 'test-user-1'. The checkbox 'Provide user access to the AWS Management Console - optional' is checked. Under 'Console password', the 'Custom password' option is selected. A 'Show password' checkbox is also present. At the bottom of the section, there is a checkbox for 'Users must create a new password at next sign-in - Recommended'. The footer of the console shows the AWS logo, search bar, and navigation links.

Step 1
● Specify user details
● Set permissions
○ Review and create
○ Retrieve password

Specify user details

User details

User name
test-user-1
The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and + = , @ _ - (hyphen)

☒ Provide user access to the AWS Management Console - optional
In addition to console access, users with `SignInLocalDevelopmentAccess` permissions can use the same console credentials for programmatic access without the need for access keys.

Console password

☐ Autogenerated password
You can view the password after you create the user.

☒ Custom password
Enter a custom password for the user.

☐ Show password

☐ Users must create a new password at next sign-in - Recommended
Users automatically get the [IAMUserChangePassword](#) policy to allow them to change their own password.

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Step 1
● Specify user details
● Set permissions
● Review and create
○ Retrieve password

Review and create

Review your choices. After you create the user, you can view and download the autogenerated password, if enabled.

User details

User name test-user-1	Console password type Custom password	Require password reset No
--------------------------	--	------------------------------

Permissions summary

Name	Type	Used as
AmazonEC2FullAccess	AWS managed	Permissions policy
AmazonS3FullAccess	AWS managed	Permissions policy

Tags - optional
Tags are key-value pairs you can add to AWS resources to help identify, organize, or search for resources. Choose any tags you want to associate with this user.
No tags associated with the resource.

aws

Search

[Alt+S]

Global

Poojith R (956035085469)

Poojith R

IAM > IAM users > Create user

User created successfully

You can view and download the user's password and email instructions for signing in to the AWS Management Console.

View user

Step 1

Specify user details

Step 2

Set permissions

Step 3

Review and create

Step 4

Retrieve password

Retrieve password

You can view and download the user's password below or email users instructions for signing in to the AWS Management Console. This is the only time you can view and download this password.

Console sign-in details

Email sign-in instructions

Console sign-in URL

https://956035085469.signin.aws.amazon.com/console

User name

test-user-1

Console password

Show

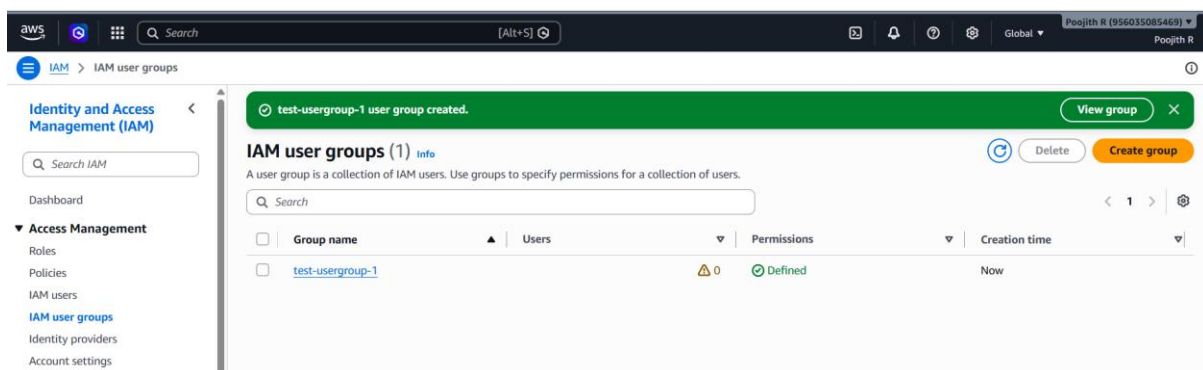
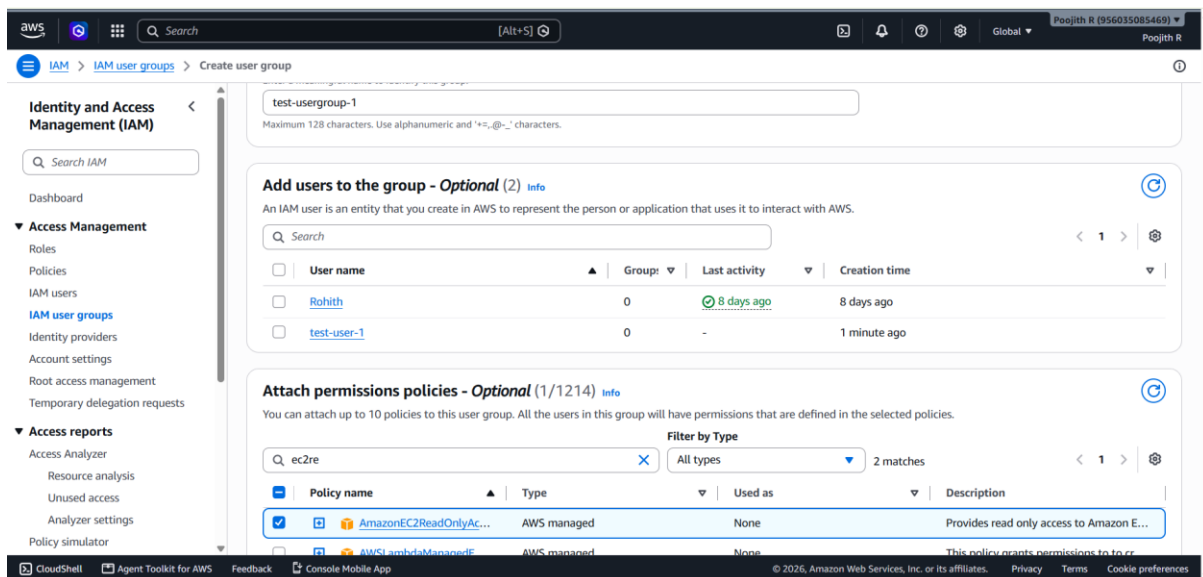
Cancel

Download .csv file

Return to users list

2. Create one group in IAM and assign read access for EC2.

- ➔ Open IAM → User groups.
- ➔ Click Create group.
- ➔ Enter the group name.
- ➔ Search for the AmazonEC2ReadOnlyAccess policy.
- ➔ Select the policy.
- ➔ Create the group.
- ➔ Verify that the AmazonEC2ReadOnlyAccess policy is attached to the group.
- ➔ Users added to this group will automatically receive EC2 read-only permissions



3. Create a new user named "Devops" and add to the group created in task 2.

- ➔ Open IAM → Users → Create user.
- ➔ Enter the username as Devops.
- ➔ Create the user.
- ➔ Open the Devops user.
- ➔ Select Add permissions → Add user to group.
- ➔ Select the EC2ReadonlyGroup.
- ➔ Add the user to the group.

- ➔ Verify that Devops is listed as a member of the group.
- ➔ Test the user and verify that EC2 resources can be viewed but cannot be modified

The screenshot shows the 'Set permissions' step in the AWS IAM console. The left sidebar indicates the current step is 'Set permissions' (Step 2 of 4). The main content area is titled 'Set permissions' and includes a sub-header 'Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. [Learn more](#)

Under 'Permissions options', three radio buttons are present: 'Add user to group' (selected), 'Copy permissions', and 'Attach policies directly'. The 'Add user to group' option has a description: 'Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.'

Below this, the 'User groups (1/1)' section shows a table with one group:

Group name	Users	Attached policies	Created
test-usergroup-1	0	AmazonEC2ReadOnlyAccess	2026-09-03 (1 minute ago)

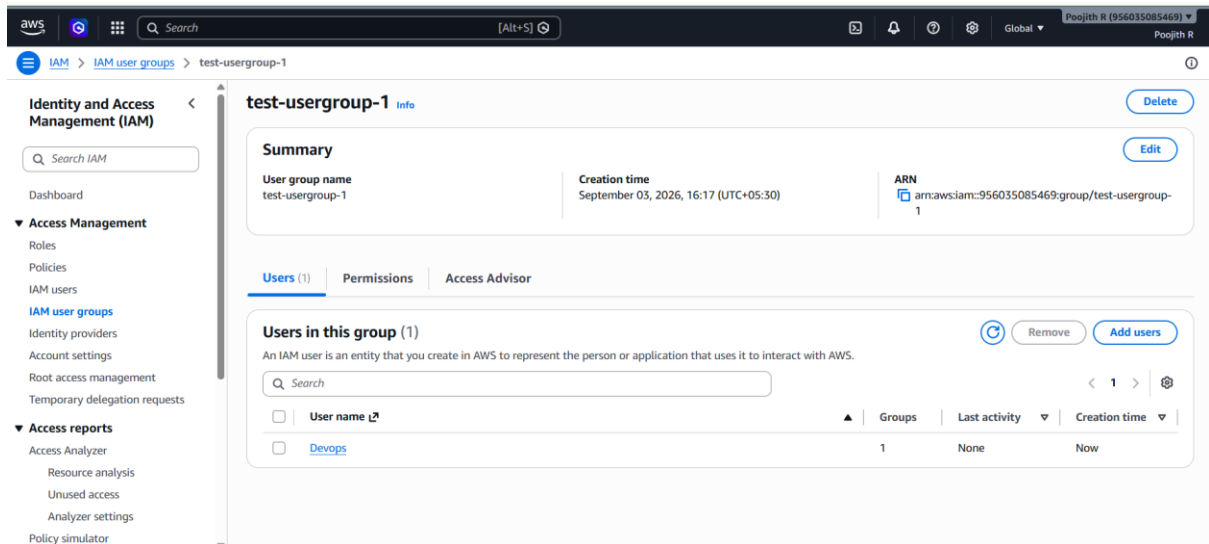
At the bottom right, there are 'Cancel', 'Previous', and 'Next' buttons.

The screenshot shows the 'Retrieve password' step in the AWS IAM console. The left sidebar indicates the current step is 'Retrieve password' (Step 4 of 4). A green success message at the top states: 'User created successfully. You can view and download the user's password and email instructions for signing in to the AWS Management Console. [View user](#)

The main content area is titled 'Retrieve password' and includes a sub-header 'You can view and download the user's password below or email users instructions for signing in to the AWS Management Console. This is the only time you can view and download this password.'

Under 'Console sign-in details', there is a 'Console sign-in URL' field with the value: <https://956035085469.signin.aws.amazon.com/console>. There is also a 'User name' field with the value 'Devops' and a 'Console password' field with a 'Show' button.

At the bottom right, there are 'Cancel', 'Download .csv file', and 'Return to users list' buttons.



4. Write a bash script to create an IAM user with VPC full access.

- ➔ Install and configure the AWS CLI.
- ➔ Create a Bash script named create-vpc-user.sh.
- ➔ Define the IAM username inside the script.
- ➔ Use the aws iam create-user command to create the IAM user.
- ➔ Attach the AmazonVPCFullAccess policy to the user.
- ➔ Save the script.

```
#!/bin/bash
```

```
USER_NAME="Poojith"
```

```
aws iam create-user --user-name "$USER_NAME"
```

```
aws iam attach-user-policy \
```

```
--user-name "$USER_NAME" \
```

```
--policy-arn arn:aws:iam::aws:policy/AmazonVPCFullAccess
```

```
echo "IAM user $USER_NAME created Successfully"
```

```
Echo "VPC Full Access Policy attached."
```

- ➔ Give all permissions to the script: `chmod 777 test.bash`
- ➔ Create a role with IAMFullAccess policy AND then add to ec2-instance IAM role
- ➔ Run the script: `./test.bash`
- ➔ Verify the attached policy: `aws iam list-attached-user-policies --user-name VPCAdmin`
- ➔ Confirm that AmazonVPCFullAccess is attached to the newly created IAM user.

AWS Search [Alt+S] United States (N. Virginia) Poojith R (956035085469) Poojith R

EC2 > Instances

Instances (1/1) Info Last updated less than a minute ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Saved filter sets Choose filter set Find Instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Put
test-instance	i-0537c2a3d01ed0cff	Pending	t3.micro	-	us-east-1d	ec2-54-221-136-102.co...	54.

i-0537c2a3d01ed0cff (test-instance)

[Details](#) [Status and alarms](#) [Monitoring](#) [Security](#) [Networking](#) [Storage](#) [Tags](#)

Instance summary [info](#)

Instance ID Public IP address Private IP addresses

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Identity and Access Management (IAM)

Search IAM

IAM users (3) Info [Delete user](#) [Create user](#)

An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.

Search

User name	Path	Group	Last activity	MFA	Password age	Console last sign-in	Acc
Devops	/	1	-	-	38 minutes	-	-
Rohith	/	0	8 days ago	-	8 days	8 days ago	-
test-user-1	/	0	-	-	42 minutes	-	-

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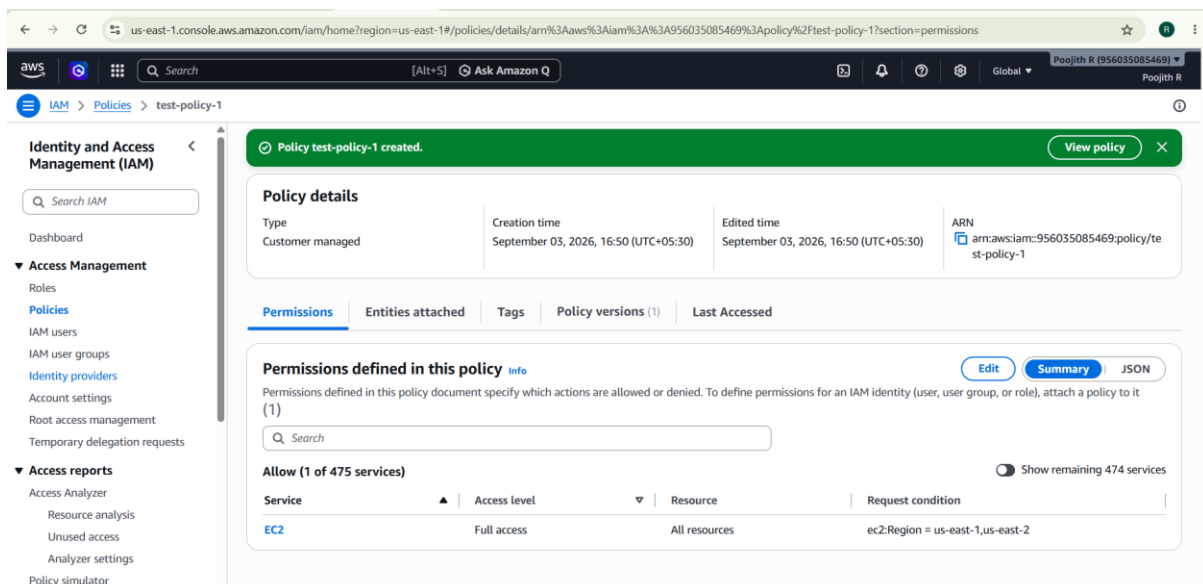
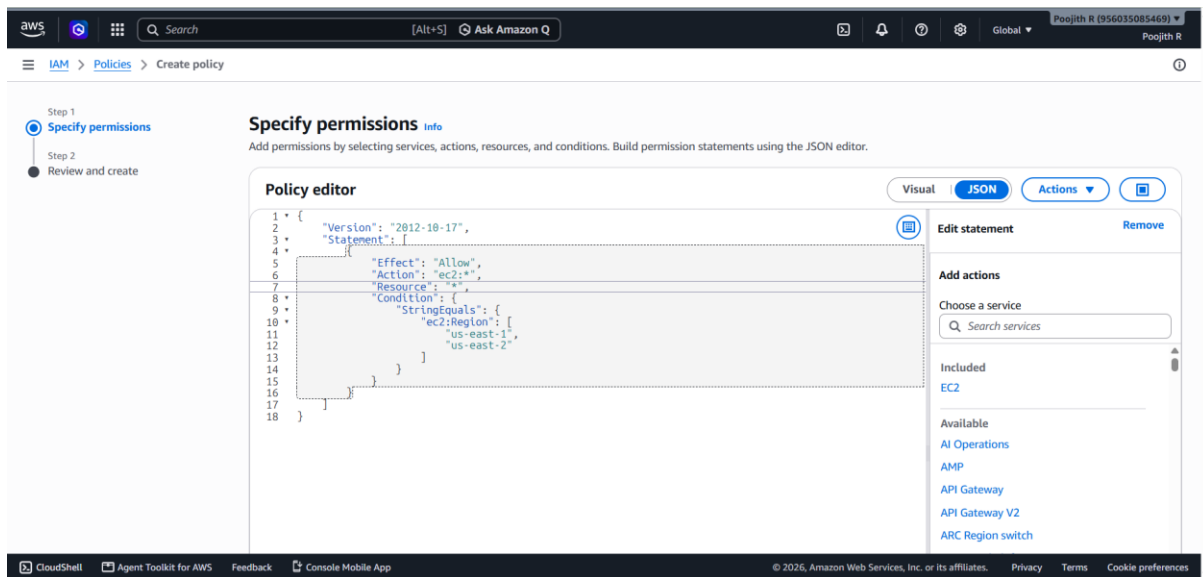
```
root@ip-172-31-36-204:~#
poojith@poojith:~/Downloads$ ssh -i Test1.pem ec2-user@54.221.136.102
Warning: Permanently added '54.221.136.102' (SSH2-0096) to the list of known hosts.
root@ip-172-31-36-204:~#
root@ip-172-31-36-204:~# cat /etc/os-release
NAME="Amazon Linux 2023"
VERSION="2023"
ID="amzn"
ID_LIKE="fedora"
VARIANT="Linux"
VARIANT_ID="linux"
VERSION_ID="2023"
PRETTY_NAME="Amazon Linux 2023"
ANSI_COLOR="0;32"
LOGO="https://aws.amazon.com/linux/amazon-linux-2023"
CPE_NAME="cpe:2.3:os:linux:amzn:2023:1:EL9"

```

```
root@ip-172-31-36-204:~# sudo ls
ls: cannot access 'ls': No such file or directory
root@ip-172-31-36-204:~# sudo su -
root@ip-172-31-36-204:~# ls
ls: cannot access 'ls': No such file or directory
root@ip-172-31-36-204:~# vi test.bash
root@ip-172-31-36-204:~# chmod 777 test.bash
root@ip-172-31-36-204:~# vi test.bash
root@ip-172-31-36-204:~# ./test.bash
Unable to locate credentials. You can configure credentials by running "aws login".
Unable to locate credentials. You can configure credentials by running "aws login".
IAM user 'poojith' created successfully.
VPC Full Access policy attached.
root@ip-172-31-36-204:~# vi test.bash
root@ip-172-31-36-204:~# aws configure
AWS Access Key ID [None]: AKIA55GBBW50SWPT33F8
AWS Secret Access Key [None]: oK5tR/fV192ok2+e2GADmw5vGZ5huJgh9FXpPw/
Default region name [None]: us-east-1
Default output format [None]: json
root@ip-172-31-36-204:~# ./test.bash
{
  "User": {
    "Path": "/",
    "UserName": "Poojith",
    "UserId": "AIDA55GBBW50SWPT33F8",
    "Arn": "arn:aws:iam::956035085469:user/Poojith",
    "CreateDate": "2026-09-03T11:32:15+00:00"
  }
}
IAM user 'poojith' created successfully.
VPC Full Access policy attached.
root@ip-172-31-36-204:~#
```

5. Create an IAM policy to allow EC2 access for a specific user in specific regions only.

- ➔ Open IAM → Policies.
- ➔ Click Create policy.
- ➔ Select the JSON policy editor.
- ➔ Define the required EC2 permissions.
- ➔ Use the ec2: Region condition to restrict access to specific AWS Regions.
- ➔ Create the policy.
- ➔ Attach the policy to the required IAM user.
- ➔ Test EC2 access in the allowed regions.



6. We have two accounts: Account A and Account B. Account A user should access an S3 bucket in Account B.

(Collaborate with a team member and execute this. This is mostly asked in every interview.)

- ➔ Account A: Contains the IAM user Test-user.
- ➔ Account B: Contains the S3 bucket.
- ➔ Account B – Configure S3 Bucket
- ➔ Log in to Account B.
- ➔ Open S3 and create or select an existing bucket.
- ➔ Note the bucket name, for example account-b-devops-bucket.
- ➔ Open the bucket and go to Permissions.
- ➔ Open Bucket Policy.
- ➔ Specify the IAM user from Account A as the principal.
- ➔ Grant the required permissions such as s3:ListBucket, s3:GetObject, and s3:PutObject.
- ➔ Save the bucket policy.

Example:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "AllowAccountAUser",
      "Effect": "Allow",
      "Principal": {
        "AWS": "arn:aws:iam::ACC_ID:user/Devops"
      },
      "Action": [
        "s3:ListBucket"
      ],
      "Resource": "arn:aws:s3:::account-b-devops-bucket-0509"
    },
    {
      "Sid": "AllowAccountAUserObjects",
      "Effect": "Allow",
      "Principal": {
        "AWS": "arn:aws:iam::acc_ID:user/Devops"
      },
      "Action": [
        "s3:GetObject",
        "s3:PutObject"
      ],
    },
  ]
}
```


"Resource": "arn:aws:s3:::account-b-devops-bucket-0509/*"

}

]

}

- ➔ Account A – Configure IAM User
- ➔ Log in to Account A.
- ➔ Open IAM → Users → Test-user.
- ➔ Attach an IAM policy that allows the required S3 actions.
- ➔ Specify the Account B S3 bucket as the resource.
- ➔ Save the policy.
- ➔ Configure AWS CLI using the Account A Devops credentials.
- ➔ Test access to the Account B bucket:
aws s3 ls s3:// account-b-devops-bucket-050910
- ➔ Upload a test file:
aws s3 cp test.txt s3:// account-b-devops-bucket-050910/
- ➔ Verify the uploaded file from Account B.
- ➔ Confirm that the Account A user can access the S3 bucket owned by Account B.

